

LAPORAN PRAKTIKUM
Aplikasi Berbasis Platform

MODUL 4-5
PRAKTIKUM WIDGET DAN LAYOUT FLUTTER



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PROGRAM STUDI S1 INFORMATIKA

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Source Code

```
import 'package:flutter/material.dart';

void main() {
  runApp(const MyApp());
}

class MyApp extends StatelessWidget {
  const MyApp({super.key});

  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      debugShowCheckedModeBanner: false,
      title: 'Tugas Praktikum Modul 4-5',
      theme: ThemeData(primarySwatch: Colors.blue),
      home: const PraktikumScreen(),
    );
  }
}

class PraktikumScreen extends StatelessWidget {
  const PraktikumScreen({super.key});

  @override
  Widget build(BuildContext context) {
    // Data array untuk ListView.builder
    final List<String> daftarPenyakit = [
      'Melanoma',
      'Karsinoma',
      'Psoriasis',
      'Dermatitis Atopik'
    ];
```

```

// Data array untuk ListView.separated
final List<String> langkahDeteksi = [
    'Ambil atau unggah foto kulit',
    'Proses pemindaian gambar (Real-Time)',
    'Tampilkan persentase akurasi & hasil'
];

return Scaffold(
    appBar: AppBar(
        title: const Text('Modul 4-5: Implementasi UI'),
        backgroundColor: Colors.teal,
        foregroundColor: Colors.white,
    ),
    // Menggunakan SingleChildScrollView agar keseluruhan halaman bisa di-scroll
    body: SingleChildScrollView(
        padding: const EdgeInsets.all(16.0),
        child: Column(
            crossAxisAlignment: CrossAxisAlignment.start,
            children: [

                // -----
                // 1. CONTAINER
                // -----
                const Text('1. Container', style: TextStyle(fontWeight: FontWeight.bold,
fontSize: 16)),
                const SizedBox(height: 8),
                Container(
                    height: 80,
                    width: double.infinity,
                    decoration: BoxDecoration(
                        color: Colors.blueAccent,
                        borderRadius: BorderRadius.circular(12),

```

```

        boxShadow: const [BoxShadow(color: Colors.black26, blurRadius: 4)]
      ),
      alignment: Alignment.center,
      child: const Text(
        'Ini adalah Container Berwarna',
        style: TextStyle(color: Colors.white, fontSize: 16, fontWeight:
FontWeight.bold),
      ),
    ),
    const SizedBox(height: 24),

    // -----
    // 2. STACK
    // -----
    const Text('2. Stack', style: TextStyle(fontWeight: FontWeight.bold,
fontSize: 16)),
    const SizedBox(height: 8),
    Stack(
      alignment: Alignment.center,
      children: [
        // Lapisan 1 (Paling bawah)
        Container(
          height: 120,
          width: double.infinity,
          decoration: BoxDecoration(
            color: Colors.grey[300],
            borderRadius: BorderRadius.circular(8),
          ),
        ),
        // Lapisan 2 (Tengah)
        Container(
          height: 80,
          width: 250,

```

```

        color: Colors.teal[300],
      ),
      // Lapisan 3 (Paling atas)
      const Text(
        'Teks Bertumpuk (Stack)',
        style: TextStyle(color: Colors.black87, fontWeight:
FontWeight.bold, fontSize: 18),
      ),
    ],
  ),
  const SizedBox(height: 24),

  // -----
  // 3. GRIDVIEW
  // -----
  const Text('3. GridView (Minimal 6 Item)', style: TextStyle(fontWeight:
FontWeight.bold, fontSize: 16)),
  const SizedBox(height: 8),
  GridView.count(
    shrinkWrap: true, // Wajib agar tidak error overflow di dalam Column
    physics: const NeverScrollableScrollPhysics(), // Mematikan scroll
bawaan grid
    crossAxisCount: 3, // Jumlah kolom
    crossAxisSpacing: 10,
    mainAxisSpacing: 10,
    children: List.generate(6, (index) {
      return Container(
        decoration: BoxDecoration(
          color: Colors.orange[(index % 9 + 1) * 100],
          borderRadius: BorderRadius.circular(8),
        ),
        alignment: Alignment.center,
        child: Text('Grid ${index + 1}', style: const
TextStyle(fontWeight: FontWeight.bold)),
      );
    }),
  ),
  const SizedBox(height: 24),

```

```

// -----
// 4. LISTVIEW (Item A, B, C)
// -----
const Text('4. ListView Dasar', style: TextStyle(fontWeight:
FontWeight.bold, fontSize: 16)),
const SizedBox(height: 8),
ListView(
  shrinkWrap: true,
  physics: const NeverScrollableScrollPhysics(),
  children: const [
    Card(child: ListTile(leading: Icon(Icons.looks_one), title:
Text('Item A'))),
    Card(child: ListTile(leading: Icon(Icons.looks_two), title:
Text('Item B'))),
    Card(child: ListTile(leading: Icon(Icons.looks_3), title: Text('Item
C'))),
  ],
),
const SizedBox(height: 24),

// -----
// 5. LISTVIEW.BUILDER (Dari Array)
// -----
const Text('5. ListView.builder (Data Array)', style:
TextStyle(fontWeight: FontWeight.bold, fontSize: 16)),
const SizedBox(height: 8),
ListView.builder(
  shrinkWrap: true,
  physics: const NeverScrollableScrollPhysics(),
  itemCount: daftarPenyakit.length,
  itemBuilder: (context, index) {
    return Card(
      color: Colors.indigo[50],
      child: ListTile(
        leading: const Icon(Icons.coronavirus, color: Colors.indigo),
        title: Text(daftarPenyakit[index], style: const
TextStyle(fontWeight: FontWeight.w500)),
      ),
    );
  },
);

```

```

        );
    },
),
const SizedBox(height: 24),

// -----
// 6. LISTVIEW.SEPARATED
// -----

const Text('6. ListView.separated (Dengan Garis)', style:
TextStyle(fontWeight: FontWeight.bold, fontSize: 16)),
const SizedBox(height: 8),
Container(
  decoration: BoxDecoration(
    border: Border.all(color: Colors.grey.shade300),
    borderRadius: BorderRadius.circular(8)
  ),
),
child: ListView.separated(
  shrinkWrap: true,
  physics: const NeverScrollableScrollPhysics(),
  itemCount: langkahDeteksi.length,
  separatorBuilder: (context, index) => const Divider(height: 1,
color: Colors.grey),
  itemBuilder: (context, index) {
    return ListTile(
      leading: CircleAvatar(
        backgroundColor: Colors.teal,
        foregroundColor: Colors.white,
        child: Text('${index + 1}')
      ),
      title: Text(langkahDeteksi[index]),
    );
  },
),
),
const SizedBox(height: 20),

],
),

```

```
        ),
    );
}
}
```

Output

Modul 4-5: Implementasi UI

1. Container

Ini adalah Container Berwarna

2. Stack

Teks Bertumpuk (Stack)

3. GridView (Minimal 5 Item)

Grid 1

Grid 2

Grid 3

Modul 4-5: Implementasi UI

4. ListView Dasar

1 Item A

2 Item B

3 Item C

5. ListView.builder (Data Array)

Melanoma

Karsinoma

Psoriasis

Dermatitis Atopik

6. ListView.separated (Dengan Garis)

1 Ambil atau unggah foto kulit

2 Proses pemindaian gambar (Real-Time)

3 Tampilkan persentase akurasi & hasil

